Data Visualization

Topic: Meteorite Landings
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School of Graduate Professional Studies
DAAN 871 - Data Visualization

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DOCUMENT CONTROL

Work carried out by:

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REVISION SHEET

Release No.	Date	Revision Description
1	7/17/2025	Part 1 of project
2	7/20/2025	Part 2 of project
3	7/27/2025	Part 3 of project
4	8/3/2025	Part 4 of project
5	8/20/2025	Final Submission

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WEEK 9 GOAL OF THE COURSE PROJECT

In this week you have to complete the following tasks

1. Define your research/business goal.
    1. Highlight interesting information about meteorite landing sites across the world to answer questions about the average frequency, size, and landing points of meteorites historically. This can provide insight into whether it is reasonable to fear for destructive meteorite landings in the future. Another goal is to see if any patterns can be identified and studied in the data, or if it more closely resembles noise.
2. Briefly describe the queries (research/business questions) you want to address. Limit the number of queries to a maximum of four.
    1. What time periods have the greatest number of recorded meteorites?
    2. What is the distribution of the weights of meteorites discovered?
    3. Are there any locations that for one reason or another seem particularly prone to meteorite landings? And have these locations changed over time?
    4. Do meteorite landings seem to happen at semi-regular intervals, or are there large time gaps between landings, followed by a period where many meteorite landings occur?
3. Briefly discuss the source(s) from which you are planning to collect the data to address your research/business queries.
    1. All data is to be sourced from the following dataset: https://catalog.data.gov/dataset/meteorite-landings. This dataset contains all relevant information on 34,513 meteorites that have been discovered and cataloged as of the writing of this report.

WEEK 10 DATA COLLECTION

Once you have identified the dataset you have to complete the following tasks:

- Provide a brief description of the dataset.
    - This dataset contains 34,513 rows corresponding to all discovered meteorites as of the writing of this report
    - Name: nominal, string, title
    - Id: nominal, numerical, identification number
    - Nametype: categorical, string, “valid” or “relict” if it has undergone excessive weathering
    - Recclass: categorical, string, unsure what this refers to
    - mass (g): numerical continuous, ratio measurement, mass to nearest gram
    - fall: categorical, string, “Fell” or “found”
    - year: numerical discrete, interval measurement, year that the meteorite fell
    - reclat: numerical continuous, interval measurement, latitude of meteorite
    - reclong: numerical continuous, interval measurement, longitude of meteorite
    - GeoLocation: string combination of longitude and latitude
- No attributes contain missing values.

WEEK 11 PURPOSE OF DASHBOARD

In this week your task will be to provide details about the purpose of the dashboard, who its users will be, what data will drive each visualization in the dashboard, a list of analytic questions and queries that a person should be able to answer using the dashboard visualizations.

Purpose of Dashboard: provide an explorative overview of the dataset to an interested general population, and give a small set of deep-dive information into answering some common questions that a general population might ask.

Audience: general population, meaning most likely American, Adult, moderate economic-class individuals who have a passing interest in science and would likely have a baseline knowledge of what a meteor is, but would be interested in learning more about meteorites and their landing sites if presented with an attractive and easy to interpret graphic which can clearly present data and answer questions are they arise.

Data: All data will be derived from the US catalog of data dataset on meteorite landings.

Analytic questions:

1. What time periods have the greatest number of recorded meteorites?
2. What is the distribution of the weights of meteorites discovered?
3. Are there any locations that for one reason or another seem particularly prone to meteorite landings? And have these locations changed over time?
4. Do meteorite landings seem to happen at semi-regular intervals, or are there large time gaps between landings, followed by a period where many meteorite landings occur?

WEEK 12 RATIONALE BEHIND THE CONTENTS OF THE DASHBOARD

In this week your task will be to provide the rationale for the different design principles and techniques you will be using in creating the interactive dashboard visualizations.

One technique I will be using is a map to show the density of discovered meteorite landings. It will include points whose size corresponds to the weight of the discovered meteorite. This is intuitive as weight is a close analogue to size, and it provides additional information. The information provided isn't directly relevant to what is being highlighted by the graphic, namely the more dense areas, but it is still helpful to show. The map will be interactive, and the user will be able to move it around and zoom into areas of interest to them to answer questions such as if there have been discoveries near where they live or grew up.

Another visualization technique will be a histogram of meteorite masses that can be used to visualize how the distribution of masses looks. This lets someone get more context about whether a certain meteorite of interest that may have been discovered near them was a particularly large or small meteorite. It also gives context on any future discoveries or news stories one may hear in the future. This is an important part of the story being conveyed. An interactive component may be the ability to specify subdivisions to adjust the display on the fly.

The third graphic will be another histogram of meteorite landings over time. This is again to give context to how frequently we would expect meteorite landings and will have similar characteristics to the previous graphic.

Finally, there will be a graphic highlighting some extremes of interest, such as if there are any remarkably large, or remarkably old meteorites in the dataset that would be of general interest to anyone using the app and would allow for that information to be conveyed without specifically looking for it in the interactive features of the previous graphics. This would be in the form of a table.

WEEK 13 FINAL SUBMISSION

This Qlik Sense dashboard is used to visualize a dataset on several statistics from the US National Aeronautics and Space Administration (NASA). As a large and public dataset, a use case is assumed for an average adult living in the US who has a passing interest in science, and the dashboard is designed to answer common questions one such individual may have, including interactive ways to manipulate the dashboard in order to provide additional clarity. This dashboard includes 2 sheets for the visualization.

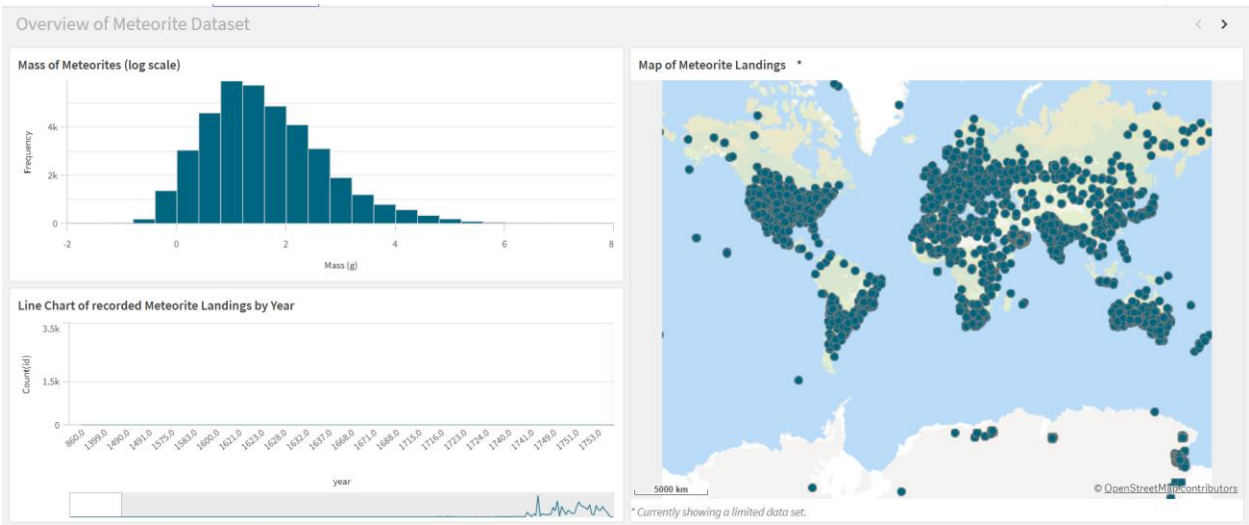

Fig 1. Sheet 1 without any selections applied

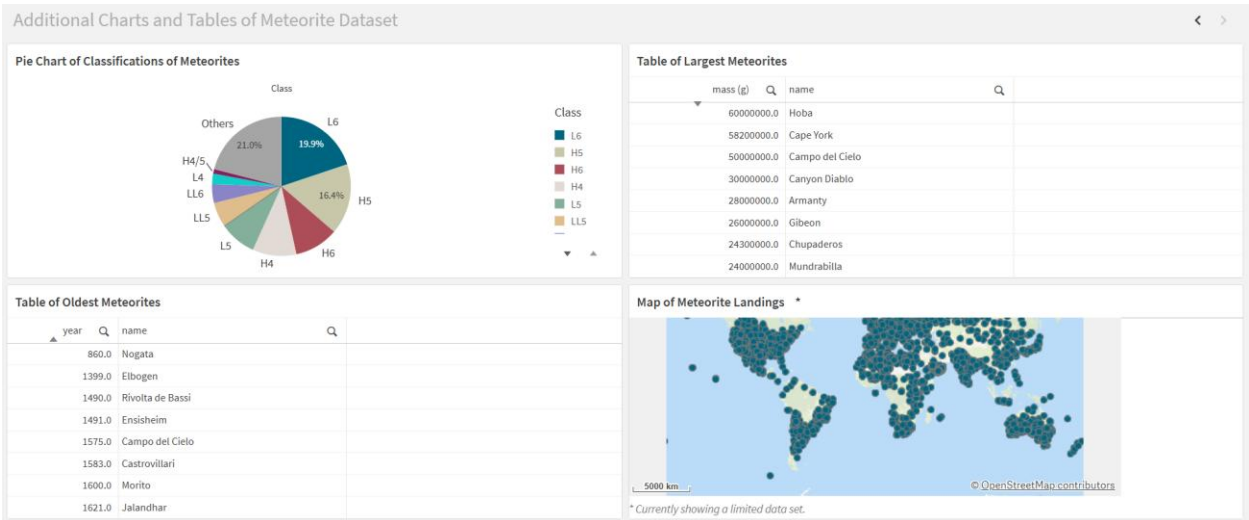

Fig 2. Sheet 2 without any selections applied

The map in both sheets is identical, and allows for interactive visualization without the need to change sheets. The histogram is in a log (base 10) scale in order to convey the distribution of mass across many orders of magnitude. The classification pie chart in the second sheet is somewhat more technical information that will most likely not be of interest to the average user of the dashboard, however it is not taking away from the presentation of the other data or causing

clutter or confusion, so it is included for completeness in order to fully display the data present in the dataset.

The specific use case explored will be a hypothetical user at Penn State interested in which landing is closest to University Park.

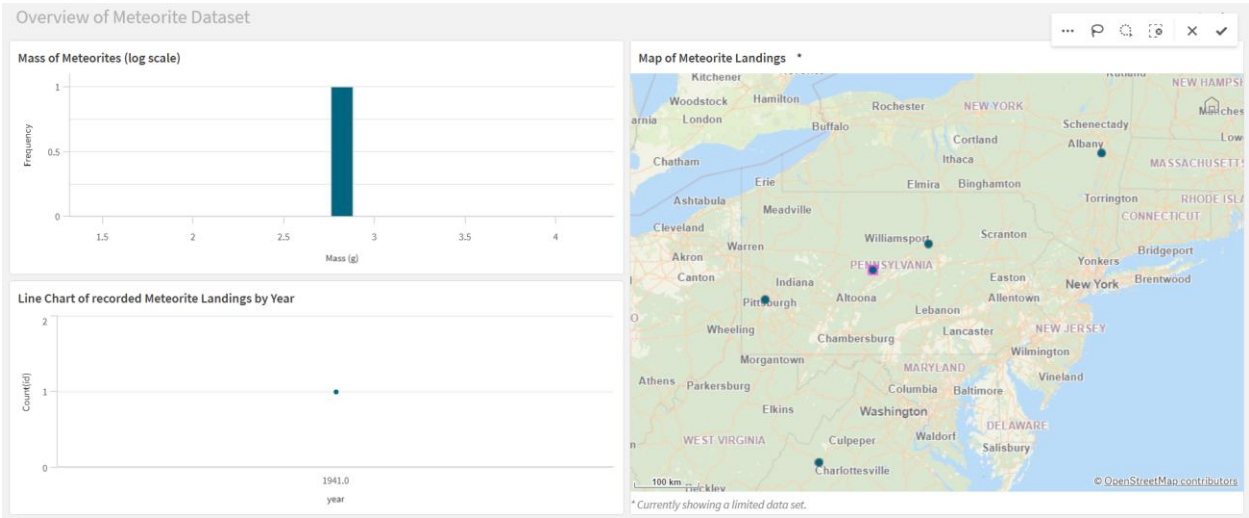

Fig 3. Sheet 1 with selection of nearest landing to University Park applied

From sheet 1, we can zoom in to the map and find the marked location closest to University Park, and select it to get information on the mass and date associated with it. We can also see that it is on the heavier end of meteorites in the database. These observations could easily be applied to any other point in the dataset, assuming one has the ability to navigate a world map to select a point in order to get information on it.

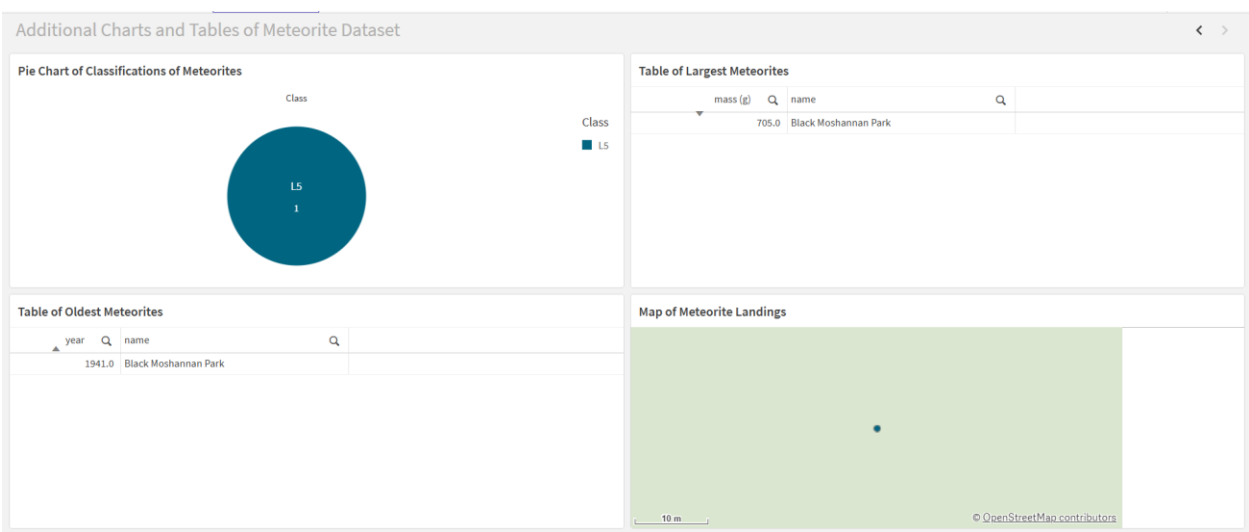

Fig 4. Sheet 1 with the selection of nearest landing to University Park applied

Keeping the same selection and going to sheet 2, we can get additional information, such as the assigned name, exact mass, and classification of the meteorite. So we can see that its name is Black Moshannan Park, its mass is 705 g, and it is classified as L5. From this we can see that selections on one sheet transfer to the other allowing for more information to be gotten from a single selection. Additionally the reverse is true and the sorting functionality on the second sheet can be used to make a selection of interest that can be used to get information from the first sheet as well. This functionality will be demonstrated later.

Another potential question of interest is how many have been reported over more recent years.

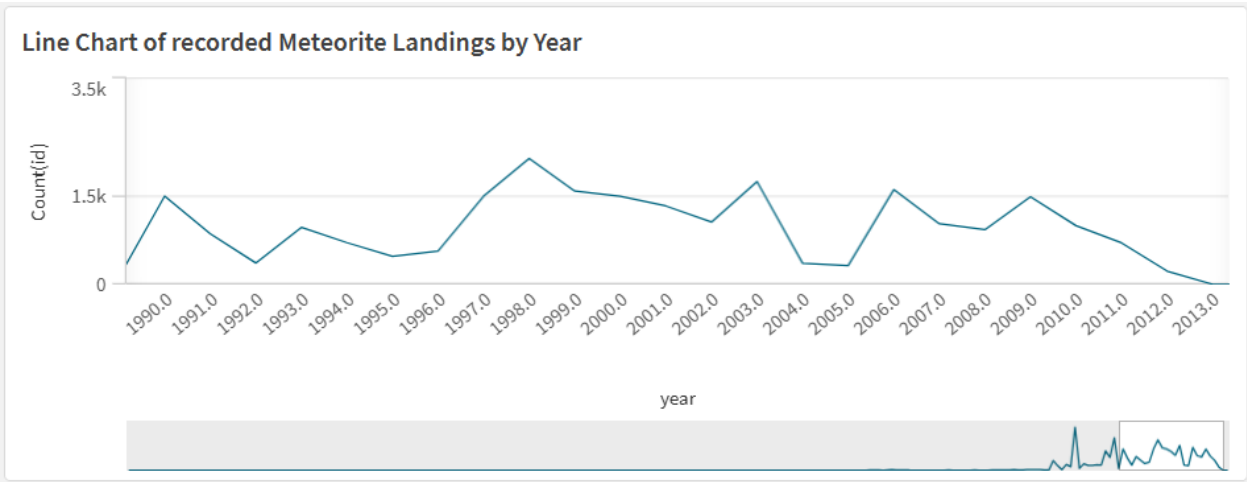

Fig 5. Line Chart from Sheet 1 having been shifted over to the most recent data in the dataset.

From this we can see the most recent data is from 2013, and there are usually between 1000 and 2000 recorded for each year, with a noticable dip in 2004 and 2005.

Another general question that our hypothetical user may ask would be where the largest meteorite discovered would be found.

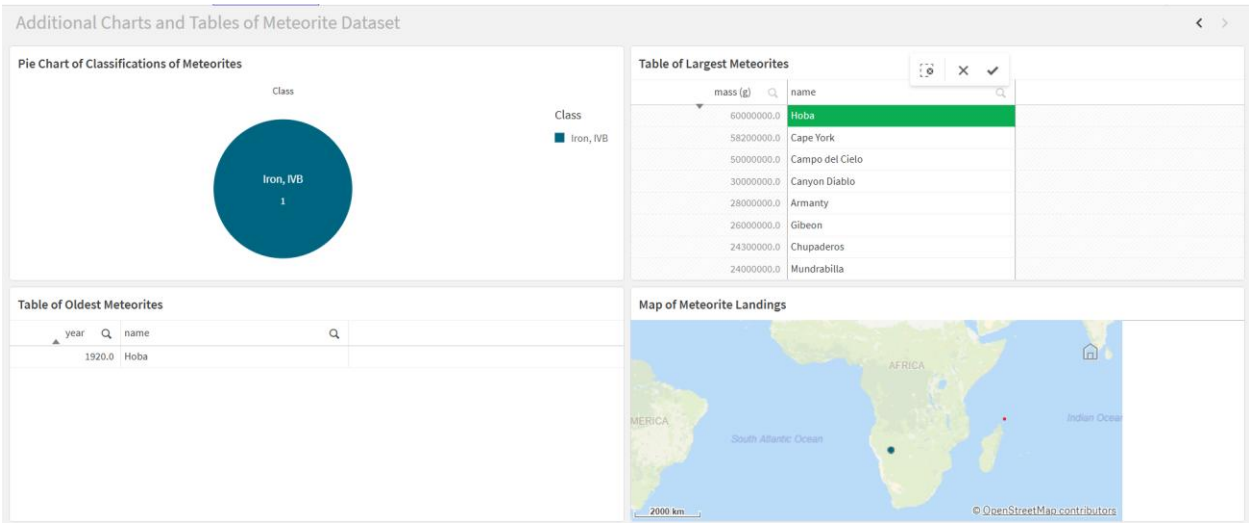

Fig 6. Sheet 2 with largest meteorite selected

By using the table, we can easily select the largest meteorite and see its year, classification, and location. It is truly massive, and given the roundness of the number, its mass is an obvious estimate.

Finally another obvious question would be information about the oldest meteorite in the dataset.

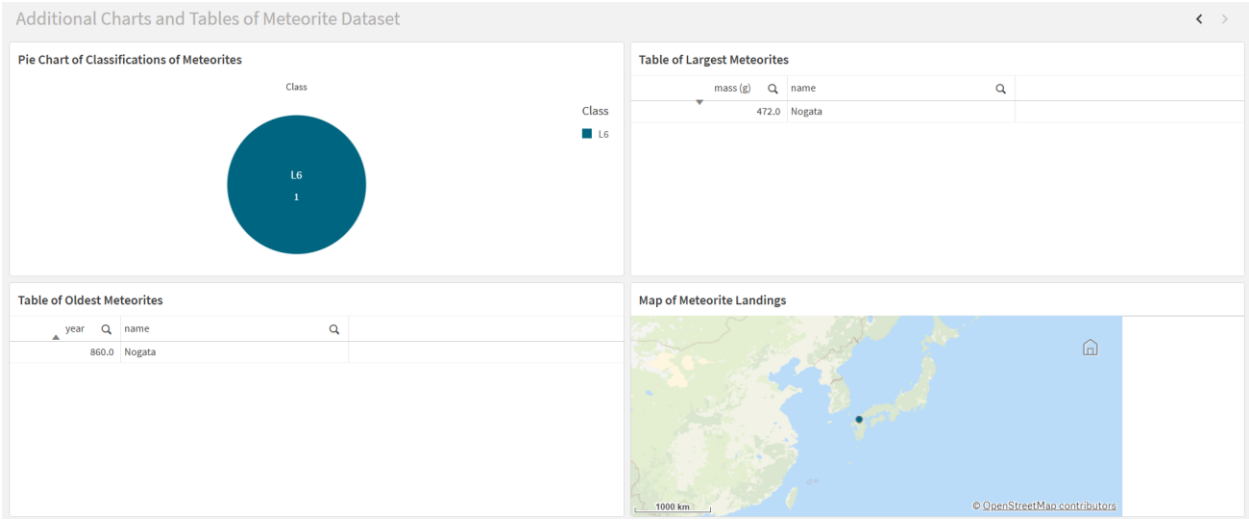

Fig 7. Sheet 2 with the oldest meteorite selected

From this selection we can see it is in the south of Japan and was dated to the year 860.

This concludes the demonstation of the capabilities of the dashboard to answer the kinds of queries anticipated from the target audience.

Sources:

Dataset found at: https://catalog.data.gov/dataset/meteorite-landings

Dashboard: https://x1ceea2bejkkuke.us.qlikcloud.com/sense/app/f407569b-1e67-47a9-9c4e-3dd102906949